Examiners' Report
Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 01 Core Pure Mathematics

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General Comments

This paper provided learners with opportunities to demonstrate their knowledge and understanding across the ability range. The earlier questions were generally more accessible, while Questions 7 to 9 proved to be the most demanding.

Learners should be reminded of the importance of using the method requested in the question. In Question 8(c), the required approach was unfamiliar to many learners, and a range of incorrect methods was seen. Some learners spent a disproportionate amount of time on this part, often without gaining credit.

Most learners appeared to have sufficient time to complete the paper. In some cases, learners produced more than one solution after recognising that their initial approach had gone wrong. This was positive where a clear restart was made, although multiple unresolved attempts sometimes made the final method difficult to follow.

Comments on Individual Questions

Question 1

This question provided an accessible start to the paper.

Part (i)

The sum of the roots and the sum of the products of pairs of roots were usually found correctly. The formula for the sum of the squares of the roots was well known, although occasional sign errors or omission of the factor of 2 prevented some learners from gaining full marks.

Part (ii)

This part was also well answered. The most common approach was to find the product of the roots first, put the given expression over a common denominator of $\alpha\beta\gamma$, and then substitute the values for the product of the roots and the sum of the products of pairs of roots. Common errors included omitting the factor of 3 and substituting values in the wrong places. An alternative method using the transformation $x = \frac{3}{w}$ was rarely seen.

Part (iii)

This part proved to be more challenging. The most common approach was to attempt to expand $(1-\alpha)(1-\beta)(1-\gamma)$ and then substitute the sum of roots, the sum of product pairs of roots and the product of roots. The most common errors seen here were slips in signs in the expansion and also in the substitution.

An alternative method involving the transformation $x = 1 - w$ was rarely seen. Even less common was the approach of using $f(x) = A(1-\alpha)(1-\beta)(1-\gamma)$ where $f(x)$ was the given cubic, then finding $\frac{f(1)}{2}$ to obtain the required answer.

Question 2

Part (a)

This part was well answered. Most learners were able to expand the brackets correctly, substitute the appropriate formulae into their expansion, and factorise the $\frac{n}{3}$ or $\frac{n}{6}$ term before proceeding to the required result.

The most common errors arose from incorrect factorisation or collection of terms. These errors often led to an incorrect n term in at least one of the brackets.

Part (b)

This part produced a mixed response. Most learners were able to form a quadratic or cubic equation using their answer to part (a). They then usually found the roots using their calculator, although the strength of the conclusion offered was more variable. Many learners did not conclude that k was not an integer, or give an equivalent statement.

Question 3

Most learners found this question accessible and achieved several marks. A variety of approaches were seen, so parts (a) and (b) were marked together.

Part (a) and (b)

Parts (a) and (b) were well answered.

Most learners were able to state the second complex root. The chosen method varied in responses: some learners found the equation first and then the roots, while others found the roots first and then the equation. For this reason, parts (a) and (b) were marked together.

They would then typically proceed to form a three-term quadratic in z with two known complex roots and obtain a quadratic either. This was mainly answered by writing the two linear factors and expanding, or instead by evaluating the sum and product of the two roots.

A variety of methods were used to find the linear factor of the given cubic.

An error seen at this stage was to assume that the linear factor was of the form ($z + ...$) rather than the correct ($3z + ...$).

An alternative method seen was using the product of the two known roots and the third unknown root and equating that to the product of roots of the given cubic. Here a similar error to the previous method was seen with the product used as -17 rather than $-\frac{17}{3}$.

Another method seen, but much less frequent, was of substituting either of the known roots into the given cubic, obtaining two simultaneous equations by equating real and imaginary parts, and then solving for the constants a and b.

Part (c)

Most learners realised that the new roots would be the original roots decreased by 2. A common error was misunderstanding the transformation and adding 2 to each of the roots.

Question 4

This question proved challenging for many learners, and some did not attempt it.

Part (a)

Some learners were able to begin setting up the equation of an invariant line. However, the method required learners to form suitable simultaneous equations using $y = mx + c$, and then solve for new values of m and c. This method still appeared to be unfamiliar to many learners, with few fully correct responses seen.

Learners who understood the process were able to eliminate x'and y' to obtain a linear equation in x. They could then form and solve a three-term quadratic equation in m.

Common errors affected the accuracy marks. Some learners either omitted $+c$ where it was required or included $+c$ where it was not needed when finding the invariant lines. Other learners used $y = mx$ rather than $y = mx + c$, and therefore did not gain any of the accuracy marks. Several responses gained no marks because learners applied the method required in part (b) instead.

Part (b)

Part (b) was more successfully answered with $y = x$ often seen, with or without any working.

Question 5

Nearly all learners began this question by establishing the base case, by substituting $n = 1$ into each side of the given expression. Some learners omitted the statement that the result was "true for $n = 1$", which was required for the mark to be awarded. In many cases, however, this was stated in the final conclusion and credit was awarded.

Most learners then stated the assumption that the result was true for $n = k$. Other correct responses instead gave a statement of the summation result.

Adding the $(k+1)^{\text{th}}$ term was the key step in a successful proof, with many learners adding $(k+1)(k+3)$. The form of the $(k \ + \ 1)^{\text{th}}$ term caused difficulty for some learners.

At this stage, some learners used the standard formulae for the addition series. This gained no credit, as the question was testing the principle of induction, and not an application of the formula. Another common approach was to expand all sets of brackets, rather than use factorisation more efficiently. Learners who chose this longer method often wrote down the target expression for the $(k+1)^{\text{th}}$ sum and then attempted to expand their own expression to show that the two expressions were equal. Some learners were not awarded marks because of poor bracketing in their expressions.

The final inductive statement was usually applied well, although some statements lacked precision. The most common issue was that learners did not make the inductive step clear, namely that "true for $n = k$" implies "true for $n = k+1$". This led to the last mark not being awarded.

Question 6

Part (a)

This part was answered correctly in almost all cases, with sign errors being the main reason for any marks being not awarded.

Part (b)

There was a mixture of responses. Where errors were made, they were often in forming the Cartesian equation for the half-line. A number of learners also gave arguments of $\pm\frac{\pi}{4}$ rather than $\frac{3\pi}{4}$.

Part (c)

Most learners correctly formed the Cartesian equations of the half-line and the circle and then solved them simultaneously to find x and y. Sign errors were the main reason for any marks not being awarded in this part. Some learners also omitted the imaginary component from their final answer.

Part (d)

Most learners did not produce a fully correct solution to this part. They were generally able to describe the circle correctly, and many were able to form one part of the inequality for $\arg(z-3+2\mathrm{i})$, but not both parts.
A common error was to give the complete argument as $\frac{3\pi}{4} \leq \arg(z-3+2i) \leq 0$ or $0 \leq \arg(z-3+2i) \leq \frac{3\pi}{4}$.

Question 7

Many learners found this question demanding and made little progress in parts (c), (d) and (e). This was most evident in learners who did not gain marks in (b) and then did not fully understand what the remainder of the question was asking.

Part (a)

The method of how to find the determinant of a 3×3 matrix was well understood. Most learners knew the condition for singularity to obtain a value for k . Slips in signs, especially in the minor determinants, led to marks not being awarded.

Part (b)

This required the finding of the inverse of the given 3×3 matrix. As one of the elements of the matrix was a constant k, this required finding the inverse using a manual method rather than directly using a calculator.

Some learners did not make progress with their method to find the inverse, while others made numerical errors in otherwise correct solutions. Alternating signs in the cofactor matrix caused difficulties in some cases. Many learners did not calculate the inverse correctly, with a combination of incorrect method and poor arithmetic often leading to method marks being awarded but no accuracy marks.

Part (c)

The most common successful approach was to replace k with the value 5 and match the matrix equation to the given simultaneous equations. A significant number of learners used $(p, -5, 7)$ instead of $(p, -5, -7)$. Learners could then use their answer to part (b) to find the required values of x, y and z.

An alternative approach was to find the inverse of the coefficient matrix using a calculator and then carry out the required matrix multiplication, without using the answer to part (b). These attempts were usually successful. It was rare to see the equations solved without using matrices, although this was an acceptable method. Fully algebraic approaches were rarely successful.

Part (d)

This part was demanding for many learners. Most did not appreciate what was required, but those who used a suitable method generally obtained both marks. Some learners attempted to substitute their answers from part (c) into the second equation to find the value of p. These attempts gained no marks, as the set of equations in this part was different and therefore had different solutions.

Successful responses usually manipulated the three simultaneous equations in pairs, obtaining pairs of equations that could be compared in order to make the set of equations consistent.

Part (e)

Understanding the geometric description in part (e) proved challenging. Many learners incorrectly stated that the planes formed a sheaf. Of those who recognised that the planes formed a prism, many did not also state that the equations were inconsistent, or that there were no solutions. It was rare to see reference to none of the planes being parallel to any of the others.

Question 8

Part (a)

This part was mostly well answered. The most common method was to form vector equations for the line and the plane, then substitute the equation of the line into the equation of the plane to find the value of the parameter and hence the coordinates of the point of intersection. Where errors were made, they were most often in forming the equation of the line, for example by using $(0,2,0)$ incorrectly as its direction vector. Some invalid attempts were also seen in which learners tried to solve Cartesian equations simultaneously.

Part (b)

Most learners were able to attempt the scalar product, and many found it correctly. Some arithmetic errors were seen when evaluating the scalar product, such as when simplifying $2 + 2 - 2$. The most common error was to leave a final answer of 74.2 degrees, which was the angle between the normal to the plane and the line.

Part (c)

Very few attempts gained any credit. Most learners tried to use results from earlier parts of the question and did not recognise that another point on l_1 was needed before finding a point on l_2. They then needed to find the direction vector between the two points on l_2, leading to the equation of the line.

Typical incorrect approaches included setting up an equation involving $\cos\theta$, based on partial knowledge of vectors, and then applying it incorrectly. This led to learners making little or no progress. In some cases, several attempts were shown, with time spent unnecessarily.

Question 9

Part (a)

Most learners substituted $x = 1$ into the equation modelling the curve, and solved the resulting quadratic equation, producing two values for the constant k. Some learners did not refer back to the model to recognise that the smaller of the two roots was required.

The value of 3 was sometimes seen which could have come from inappropriate premature rounding of the smaller root ($3.057..$) or from a misunderstanding of the model, with learners instead finding where the curve crossed the y-axis.

Part (b)

This part produced a mixed response. The B1 mark was often not awarded because dy was missing from learners' statements for the volume of a solid of revolution. The integration was usually carried out correctly, with most learners expanding the function to form a cubic before integrating.

Common errors included omitting the volume of the cylinder from the model, using an incorrect formula for the volume of the cylinder, and, following an incorrect choice in part (a), using an incorrect value for k. Sufficient evidence of the use of limits was seen more consistently than in previous years. The instruction to round the final answer to the nearest cm^3 was rarely ignored.

Part (c)

This part was well answered. Common responses referred to the surface not being smooth, the equation not being a suitable model, or the surface having possible imperfections. Some comments that did not gain credit referred to the thickness of the plastic or to possible inaccuracy in the measurements.

Part (d)

When evaluating the model, most learners realised the need to compare their answer with 48 cm^3 and then comment on whether the model was good or poor. Some learners used percentage errors to support their argument.

Part (e)

This part was less well answered, with blank responses often seen. A common incorrect refinement was to suggest that the curve should be that of a sphere, showing a misunderstanding of the reference to a cross-section. The expected response was that the curve should be a circle. Some learners gave a possible equation of the circle, which was also acceptable.